



Surveys, Experiments, and Observational Studies

Three methods used to collect data are surveys, experiments, and observational studies.

Surveys are used to collect information from a population by directly asking questions. They are administered in the form of a questionnaire or an interview. Surveys can be conducted over the phone, online, through the mail, or in person.

Experiments require defining a variable that you want to study and then randomly selecting a control group and an experimental group. Only the experimental group is manipulated in some way, and the control group receives no treatment. Both groups are monitored before and after the manipulation and any differences are analyzed.

Observational studies involve watching and recording the subjects of a study without influencing or manipulating the subjects.

Each data-collection method has strengths and weaknesses. Surveys are good for gathering data from a large population using a random sample but depend on people answering questions honestly. Experiments are controlled and useful for determining the effect of a single variable, but experimental outcomes can sometimes be affected by outside influences. Observational studies allow researchers to examine how someone or something behaves naturally, without any intervention. However, it is often difficult to randomly select the subjects who are to be observed.

Identifying Surveys, Experiments, and Observational Studies

Example of a survey: A trainer wants to know what percent of the 419 members of a gym would attend a personal training session if it were half price. The trainer randomly selects 30 gym members by first assigning each gym member a number from 001 to 419. She then uses a table of random, three-digit numbers to select the first 30 numbers from 001 to 419. She calls the 30 randomly selected gym members and records their answers.

Example of an experiment: A scientist wants to know if a new antibiotic is effective against a particular bacteria. She has 20 bacterial colonies, all with the same number of bacteria. She treats 10 colonies with the antibiotic and leaves the other 10 colonies untreated. After treatment, she counts the number of live bacteria in each colony and compares the two groups.

Example of an observational study: Researchers are watching blue whales in their natural habitat and recording their eating habits. The researchers are not manipulating the situation in any way. They are only writing down what they see.

EXAMPLE

A college student is asked a few questions to rate the effectiveness of an app he uses on his mobile device. Is he being asked to participate in an experiment, a survey, or an observational study?

SOLUTION

The student is not given any treatment, so he is not participating in an experiment. The student is also not being directly observed by a researcher, so he is not part of an observational study. Because he is asked to complete a questionnaire about the app he uses, he is being asked to participate in a survey. ■